SIMATS ENGINEERING

CSA0705 - COMPUTER NETWORKS

STUDENT NAME : BALAJI A

REGISTER NO : 1911260245

DEPARTMENT : B.TECH COMPUTER SCIENCE ENGINEERING

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FACULTY NAME : KUMARAN G

NATURE OF ASSIGNMENT : INDIVIDUAL

1. Given Values and Assumption

Item	Value
Task 1 traffic per line	Robot controllers 32 Mbps; vibration sensors 0.24 Mbps; CCTV 96 Mbps; office users 30 Mbps; total 158.24 Mbps
Task 1 plant traffic (8 lines)	1,265.92 Mbps with full CCTV streams; 536.32 Mbps if CCTV is recorded locally and only 5% of its stream is sent
Task 5 mobility values	AGV speed 2 m/s; route 1.2 km = 1,200 m; Wi-Fi access-point spacing 30 m; Wi-Fi handover 150 ms; private 5G cell spacing 300 m; 5G handover 20 ms
Command deadline	10 ms for an AGV command
Original design (Tasks 4–11)	Wired robot cell, fibre backbone, Wi-Fi for AGVs and offices, leased fibre WAN to head office, retransmission delay 40 ms (Task 8)

Table 1: Values taken from the assignment and earlier tasks.

Assumption (not given in the assignment). The assignment does not state the AGV command size or rate. This report assumes each AGV receives a 200-byte command every 2 ms, the same profile as the robot controllers in Task 1. All AGV traffic figures below are estimates based on this assumption.

A. Impact Analysis (2 Marks)

A.1 What is affected

Part of design	Affected?	Reason and effect
AGV wireless choice	Yes – changed	Task 5 showed Wi-Fi handover of 150 ms. The AGVs move to private 5G URLLC with 20 ms handover.
Wi-Fi	Partly	Wi-Fi 6 stays for office users, who do not move quickly. It no longer carries AGV commands.
Robot-cell medium (Cat6A S/FTP, 85 m)	No – retained	Robot controllers are fixed devices and stay on wired Ethernet with QoS priority.
Fibre backbone (140 m multimode, 550 m single-mode)	Yes – load	It must also carry 5G base-station traffic. Load rises to 1,305.92 Mbps (full CCTV), so 10 Gbps links are used.
Head-office WAN	Yes – load	1 Gbps is already too small for 1,265.92 Mbps; the extra AGV traffic makes the shortfall larger.
Error control	Yes	AGV commands have a 10 ms deadline, but a retransmission takes 40 ms (Task 8). Retransmission cannot be relied on for them; CRC detection, FEC and redundancy are needed (Section C.3).

Table 2: Parts of the original design affected by 5G and 50 AGVs.

A.2 Traffic recalculation with 50 AGVs

Traffic per AGV, using the assumed 200-byte command every 2 ms:

$$\text{Traffic per AGV} = (200 \times 8 \text{ bits}) \div 0.002 \text{ s} = 800{,}000 \text{ bits/s} = 0.8 \text{ Mbps}$$

$$\text{Traffic for 50 AGVs} = 50 \times 0.8 \text{ Mbps} = 40 \text{ Mbps}$$

$$\text{Updated plant traffic (full CCTV)} = 1{,}265.92 + 40 = 1{,}305.92 \text{ Mbps} \approx 1.306 \text{ Gbps}$$

$$\text{Updated plant traffic (local CCTV recording)} = 536.32 + 40 = 576.32 \text{ Mbps}$$

The two scenarios must not be mixed. The first assumes every CCTV stream is sent continuously. The second assumes cameras record locally and only 5% of the CCTV stream (4.8 Mbps instead of 96 Mbps per line) is sent.

Scenario	Plant traffic (Mbps)	Share of 1 Gbps	Share of 10 Gbps	Increase caused by AGVs
Original, full CCTV (Task 1)	1,265.92	126.6%	12.7%	–
With 50 AGVs, full CCTV	1,305.92	130.6%	13.1%	+3.2%
Original, local CCTV (Task 1)	536.32	53.6%	5.4%	–
With 50 AGVs, local CCTV	576.32	57.6%	5.8%	+7.5%

Table 3: Traffic comparison before and after adding 50 AGVs (calculated, not measured).

- **Full-stream case:** 1,305.92 Mbps is more than 1 Gbps, by 305.92 Mbps. A 1 Gbps backbone is NOT sufficient. It was already insufficient before the AGVs were added.
- **Local-recording case:** 576.32 Mbps is 57.6% of 1 Gbps on average. It fits, but leaves only 42.4% for bursts, overhead and growth.
- **Effect of AGVs on volume:** 40 Mbps is only 3.1% of 1,305.92 Mbps. The AGVs are a small load; their main impact is on delay and handover (Section B), not on bandwidth.
- **Per 5G cell:** the route of 1,200 m is covered by 1,200 ÷ 300 = 4 cells. If the AGVs were spread evenly, the average is 40 ÷ 4 = 10 Mbps per cell. Uneven spreading must be checked.

B. Test the Original Design (2 Marks)

The original design served AGVs over Wi-Fi. The handover calculation for one AGV travelling the full 1,200 m route at 2 m/s is:

$$\text{Route time} = 1{,}200\ \text{m} \div 2\ \text{m/s} = 600\ \text{s}$$

$$\text{Wi-Fi handovers} = 1{,}200\ \text{m} \div 30\ \text{m} = 40;\quad \text{interruption} = 40 \times 150\ \text{ms} = 6{,}000\ \text{ms} = 6\ \text{s}$$

$$\text{5G handovers} = 1{,}200\ \text{m} \div 300\ \text{m} = 4;\quad \text{interruption} = 4 \times 20\ \text{ms} = 80\ \text{ms}$$

$$\text{Fleet of 50 AGVs: Wi-Fi } 50 \times 40 = 2{,}000 \text{ handovers};\quad \text{5G } 50 \times 4 = 200 \text{ handovers}$$

The fleet figures are aggregate counts across all AGVs. They are not the interruption felt by one AGV: each AGV experiences only its own 40 handovers (Wi-Fi) or 4 handovers (5G). The calculation assumes one handover at each cell boundary on a one-way trip.

Quantity	Wi-Fi (original)	Private 5G	Reduction
Spacing between access points / cells	30 m	300 m	10 times larger
Time between handovers at 2 m/s	15 s	150 s	10 times longer
Handovers per AGV per route	40	4	36 fewer
Interruption per AGV per route	6 s (1.0% of 600 s)	80 ms	75 times less
Fleet handovers (50 AGVs)	2,000	200	1,800 fewer
Handover delay	150 ms	20 ms	7.5 times less
Handover delay ÷ 10 ms deadline	15 times the deadline	2 times the deadline	–
Distance travelled during one handover	2 × 0.15 = 0.30 m	2 × 0.02 = 0.04 m	–
Commands lost per handover (assumed 1 per 2 ms)	150 ÷ 2 = 75	20 ÷ 2 = 10	–

Table 4: Handover comparison for one AGV and for the fleet.

Deadline test. Wi-Fi: 150 ms > 10 ms. Private 5G: 20 ms > 10 ms. Both handover delays exceed the AGV command deadline.

Conclusion for B. The original Wi-Fi design cannot serve 50 moving AGVs: each handover is 15 times the deadline, an AGV makes 40 of them, and across the fleet there are 2,000 handovers per route, about one every 0.3 s (600 s ÷ 2,000). Private 5G is a large improvement, with 10 times fewer handovers per AGV and 75 times less interruption. However, its given 20 ms handover delay alone does not guarantee the 10 ms deadline, so 5G alone does not ensure that no command is ever missed. Extra measures are needed (Section C).

C. Adaptation or Redesign (2 Marks)

The decision is to MODIFY the original design. Table 5 gives the revised design and Figure 1 shows it.

Area	Revised design
AGVs	Private 5G URLLC with overlapping coverage and tested fast handover
Robot controllers	Wired industrial Ethernet (Cat6A S/FTP, 85 m) with QoS priority
CCTV	Wired Ethernet (PoE) with local recording
Office users	Wi-Fi 6
Sensors	Wired Ethernet, or a LoRaWAN / Zigbee gateway where cabling is difficult
Shop-floor backbone	Fibre: multimode 140 m for the office link, single-mode 550 m to the data centre, 10 Gbps
Data centre	Fibre-connected core switch and servers, plus the private 5G core and an edge controller
Head office	Leased fibre WAN (400 km), starting at 10 Gbps

Table 5: Revised network design.

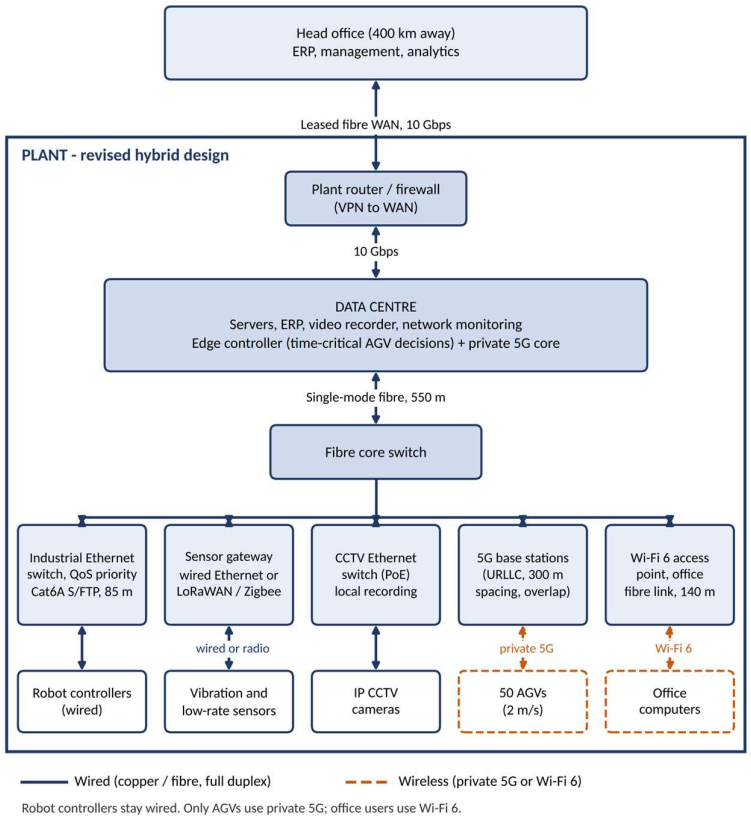


Figure 1: Revised hybrid network topology with private 5G for the AGVs.

C.1 Measures to meet the AGV deadline

These are proposed engineering improvements. They must be tested on site before they are accepted; no result is claimed here.

- **Overlapping 5G coverage:** cells overlap along the 1,200 m route, so an AGV is always in range of two base stations.
- **Make-before-break handover (where supported):** the AGV connects to the new cell before leaving the old one, so the interruption can be much shorter than 20 ms.
- **Packet duplication or redundant paths (where supported):** the same command is sent through two paths, so one lost copy does not lose the command.
- **Local edge controller:** time-critical decisions are made inside the plant, close to the AGVs, so the 10 ms limit does not depend on the 400 km WAN (2 ms one way, Task 9).
- **QoS prioritisation:** AGV and robot commands use the highest queue on the 5G and wired links.
- **Fail-safe behaviour:** if a command is missed, the AGV slows or stops safely. The network alone cannot promise zero missed commands, so safety must not depend on it alone.
- **Redundancy:** redundant power and network connections for base stations, the core switch and the edge controller where practical.
- **Monitoring:** continuous measurement of latency, jitter, packet loss and handover performance, with alarms.

C.2 Backbone and WAN

With 1,305.92 Mbps of average traffic, 10 Gbps fibre links use 13.1% (full CCTV) or 5.8% (local CCTV) of capacity. The leased WAN also starts at 10 Gbps. Peak traffic, protocol overhead and growth must still be checked.

C.3 Error control and flow control

Traffic	Error control	Flow control / reliability
AGV and robot commands (10 ms deadline)	CRC/FCS detects corrupted frames on Ethernet. FEC (channel coding) where the wireless link supports it, so small errors are corrected without waiting.	No dependence on retransmission: a retransmission takes 40 ms (Task 8), 4 times the deadline. Use packet duplication, cyclic command updates, QoS priority and a safe stop.
CCTV and sensor data	CRC detects errors; damaged frames are dropped. Local recording keeps full footage.	Rate limited by the encoder and QoS. Lost live frames are not resent.
Office, ERP and file transfer	CRC/FCS plus ARQ: retransmission of lost data.	Reliable TCP with sliding-window flow control. Congestion control is a separate TCP function.

Table 6: Error-control and flow-control choices.

Trade-off. Reliability, delay and overhead pull in different directions. ARQ gives complete data but adds delay, which suits office traffic but not 10 ms commands. FEC and duplication avoid waiting but add overhead; for example, the Hamming code for 16 data bits needs 5 check bits, an overhead of 5 ÷ 16 = 31.25% (Task 8), and duplication doubles the command traffic. For 50 AGVs the doubled command traffic would be 2 × 40 = 80 Mbps, which is small compared with a 10 Gbps backbone.

D. Evidence-Based Defence (2 Marks)

Final decision: MODIFY the original design. The original wired parts and Wi-Fi 6 for offices still work. Only the AGV wireless choice must change, and the backbone and WAN must be enlarged. A complete redesign is not needed, and keeping the original unchanged fails the AGV test.

Evidence	Original (Wi-Fi)	Revised (5G)	Meaning
Extra AGV traffic (assumed)	40 Mbps	40 Mbps	Small load (3.1%)
Plant traffic, full CCTV	1,305.92 Mbps	1,305.92 Mbps	Above 1 Gbps; 10 Gbps needed
Plant traffic, local CCTV	576.32 Mbps	576.32 Mbps	57.6% of 1 Gbps
Handovers per AGV per route	40	4	10 times fewer
Interruption per route	6 s	80 ms	75 times less
Handover delay vs 10 ms deadline	150 ms (15 times)	20 ms (2 times)	5G better, still above deadline
Fleet handovers (50 AGVs)	2,000	200	Less load on the network

Table 7: Numerical evidence for the decision.

Area	Original design	Revised design	Decision
AGVs	Wi-Fi	Private 5G URLLC, overlap, edge control, redundancy	Modify
Robot controllers	Wired Ethernet	Wired industrial Ethernet with QoS	Retain
CCTV	Wired Ethernet	Wired with local recording	Retain
Office users	Wi-Fi	Wi-Fi 6	Retain
Backbone and WAN	Fibre, 1 Gbps link in question	Fibre, 10 Gbps	Modify
Error control	CRC, ARQ	CRC, FEC, duplication for commands; ARQ for office	Modify

Table 8: Original design compared with the revised design.

Advantages, disadvantages and trade-offs

- **Advantages:** 10 times fewer handovers and 75 times less interruption per AGV; mobility designed for moving vehicles; robots and CCTV stay on proven wired links; room to add more plants and AGVs.
- **Disadvantages:** 20 ms handover is still above 10 ms until fast handover is proven; 5G needs more skills and testing; more equipment to maintain.
- **Costs:** private 5G core, base stations, SIMs and AGV modems; edge controller; 10 Gbps fibre and WAN service; redundant power and links. Wi-Fi alone would be cheaper but fails the test in Section B.
- **Implementation trade-offs:** more overlap and redundancy cost more but reduce missed commands; local recording saves WAN bandwidth but needs storage at the plant.

Conclusion

Adding 50 AGVs raises plant traffic by an estimated 40 Mbps, from 1,265.92 to 1,305.92 Mbps with full CCTV, or from 536.32 to 576.32 Mbps with local CCTV recording. The bigger problem is mobility: over a 1,200 m route Wi-Fi gives 40 handovers and 6 s of interruption per AGV, while private 5G gives 4 handovers and 80 ms. Even so, 20 ms is above the 10 ms deadline. The original design is therefore modified: private 5G URLLC for AGVs, edge control, overlapping coverage, redundancy and monitoring, with wired Ethernet, fibre and Wi-Fi 6 kept for the rest. The numbers are calculated from the stated assumption; real latency, handover and packet loss must be measured before the design is accepted.